Rank-optimal directional Taylor jets for high-order mixed derivatives of neural surrogates

Abstract

High-order partial differential equation losses often require repeated evaluation of one prescribed mixed derivative of a neural surrogate. We develop a deterministic, algebraically exact procedure for this task. A mixed derivative is first identified with a coefficient functional of the homogeneous polynomial formed by directional Taylor coefficients. Constructing a shortest directional formula is therefore equivalent to decomposing the associated monomial into powers of linear forms. The Carlini–Catalisano–Geramita rank formula gives the minimum number of complex directions, and a roots-of-unity construction attains that minimum without solving a linear system. We evaluate each direction by Taylor-mode propagation through a multilayer perceptron and use a custom vector–Jacobian product so that derivatives of the loss with respect to network parameters remain available. Double-precision tests agree with nested automatic differentiation for both values and parameter gradients. As an application, we train complex-parameter sinh physics-informed neural networks on 12 settings drawn from polyharmonic, radial-chirp, and time-harmonic Maxwell problems. Under a common 1200 s, five-seed, width-128 protocol, this model has the lowest mean relative L^2 error in all 12 settings, although the margin is negligible for the hardest three-dimensional sixth-order case. The comparison is deliberately protocol-specific because trainable real parameter counts differ. The rank reduction applies to a fixed repeated-index derivative; it gives no global optimality result for general operator sums and no reduction for square-free mixed derivatives.

Keywords: Physics-informed neural network, high-order derivative, Taylor-mode automatic differentiation, Waring decomposition, apolarity, complex-valued neural network

1. Introduction

1.1. Motivation and related work

Scientific machine-learning models frequently require derivatives of a neural surrogate with respect to its inputs. In a physics-informed neural network (PINN) [1], for example, each optimization step evaluates the differential operator in the governing equation at many collocation points. Fourth- and sixth-order equations can therefore make input differentiation a substantial part of the training cost.

A direct implementation applies reverse-mode automatic differentiation successively, retaining the graph after every coordinate derivative. The depth and storage of this nested graph grow with both the derivative order and the network depth. Three complementary strategies reduce this cost:

(i) **Forward- and Taylor-mode automatic differentiation** [2, 3], which propagates truncated Taylor coefficients through each primitive without nesting input-gradient graphs;

(ii) **Stochastic estimators** [4, 5], which write the target operator as an expectation over random tangent vectors and approximate it by sampling; and

(iii) **Operator-specific specializations** [6, 7] for the Laplacian (the trace of the Hessian), exploiting the fact that the trace collapses cleanly through linear layers.

Taylor mode supplies a derivative along any chosen direction, but does not specify a shortest set of directions for extracting one mixed partial. Stochastic methods trade work for variance, whereas trace-specialized methods address a different operator structure. Here we consider the deterministic fixed-multi-index problem and identify its directional schedule with the Waring decomposition of a monomial.

1.2. Contributions and scope

1. We prove that a directional formula for a fixed mixed derivative exists with R probes if and only if the associated degree-p monomial has a Waring decomposition with R terms. Thus the minimum probe count is a Waring rank (Section 3).

2. The Carlini–Catalisano–Geramita theorem [8] and a roots-of-unity filter give a closed-form schedule that attains the complex rank. For the repeated-index patterns $(4, 2)$ and $(2, 2, 2)$ this schedule uses 5 and 9 directions, against 6 and 13 for fully projectively consolidated real polarization; the simpler coalescing rule that we implement uses 7 directions for $(4, 2)$.

3. We combine this schedule with Taylor-mode propagation through smooth multilayer perceptrons and a custom vector–Jacobian product. The resulting implementation is algebraically exact, differentiable with respect to model parameters, and verified against nested automatic differentiation (Sections 4 and 5).

4. We apply the implementation to complex-parameter sinh PINNs and report five-seed, fixed-budget comparisons on polyharmonic, radial-chirp, and time-harmonic Maxwell problems (Section 6). The study records accuracy, throughput, and trainable real degrees of freedom rather than equating a common hidden width with parameter-count matching.

The rank statement is deliberately narrow. It concerns one monomial differential operator at a time. A sum such as Δ^m can be expanded and evaluated term by term; the schedule may shorten individual repeated mixed terms, but it does not exploit cancellations or shared structure across the sum. A trace-collapsing method may therefore be preferable when only a Laplacian is required. Square-free derivatives receive no rank reduction, and we make no claim of a universal wall-clock advantage over real polarization. Stochastic estimators remain preferable when a controlled approximation of a large tensor contraction is sufficient.

The remainder of the paper is organized as follows. Section 2 writes one fixed mixed partial derivative as a coefficient functional of the polynomial generated by directional Taylor coefficients. Section 3 proves the equivalence with monomial Waring decomposition, states the rank formula, and derives the roots-of-unity schedule and the resulting pattern taxonomy. Section 4 gives the Taylor-mode jet rules, the implemented backends, the end-to-end algorithm, and its cost. Section 5 reports the verification of the schedule, of derivative values, and of parameter gradients against nested automatic differentiation. Section 6 applies the backend to complex-parameter sinh PINNs on the polyharmonic, radial-chirp, and time-harmonic Maxwell benchmarks. Section 7 discusses scope and limitations, and Section 8 concludes. Appendix

A collects the proofs of the roots-of-unity identities and the apolar form of the rank bound.

1.3. Notation

Throughout the paper, $u : \mathbb{R}^d \to \mathbb{R}$ denotes a sufficiently smooth scalar field. Let $\nu = (\nu_1, \ldots, \nu_d) \in \mathbb{Z}^d_{\geq 0}$ be a multi-index of order $p = |\nu|$, with

$$\partial^\nu \;:=\; \partial_1^{\nu_1} \cdots \partial_d^{\nu_d}. \tag{1}$$

The implementation represents the same operator by an expanded coordinate list in which coordinate k occurs ν_k times.

Relabel the *active coordinates* $\{k : \nu_k > 0\}$ as $i_0, \ldots, i_n$, and set $a_j = \nu_{i_j}$. We order them so that $1 \leq a_0 \leq \cdots \leq a_n$ (ties may be broken by the original coordinate index). This relabeling singles out a minimum-exponent variable i_0, as required by the rank formula. Then

$$p \;=\; \sum_{j=0}^{n} a_j, \qquad \nu! \;=\; \prod_{k=1}^{d} \nu_k! \;=\; \prod_{j=0}^{n} a_j!, \qquad m_j \;=\; a_j + 1. \tag{2}$$

The *active subspace* V is the linear span of the active coordinate directions,

$$V \;:=\; \operatorname{span}\{e_{i_0}, \ldots, e_{i_n}\} \;\subset\; \mathbb{R}^d; \tag{3}$$

since ∂^ν is insensitive to coordinates outside the active set, all directional probes v below may be taken in V.

The directional Taylor coefficient of order p at x along $v \in V$ is

$$T_p(x; v) \;:=\; \frac{1}{p!} \frac{d^p}{d\tau^p} u(x + \tau v)\Big|_{\tau=0} \;=\; \frac{1}{p!} D^p u(x)[v, \ldots, v]. \tag{4}$$

For a complex direction $v \in V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$, we define $T_p(x; v)$ by the complex multilinear extension of $D^p u(x)$ in the right-hand side of (4). If u admits a holomorphic complexification $u_{\mathbb{C}}$, this definition equals the order-p coefficient of $u_{\mathbb{C}}(x + \tau v)$. This distinction is needed because an arbitrary C^p real function need not be defined at complex inputs. Finally, $\mathbb{K}[z]_p$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, denotes the space of degree-p homogeneous polynomials in $z = (z_0, \ldots, z_n)$.

2. Single-monomial partials as polynomial coefficient extraction

2.1. The generating polynomial

For $z = (z_0, \dots, z_n) \in \mathbb{R}^{n+1}$ parametrizing the active subspace via $v = \sum_j z_j e_{i_j}$, the *generating polynomial* of u at x is

$$Q_u(z) \;:=\; T_p\Big(x;\ \sum_{j=0}^{n} z_j e_{i_j}\Big), \qquad z \in \mathbb{R}^{n+1}. \tag{5}$$

By construction Q_u is a degree-p homogeneous polynomial in z; the next lemma identifies its coefficients with the partial derivatives of u.

Lemma 2.1 (Generating polynomial expansion)**.** *For $u \in C^p$ on a neighborhood of x,*

$$Q_u(z) \;=\; \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!}\, z^\beta, \tag{6}$$

where $\partial^\beta = \partial_{i_0}^{\beta_0} \cdots \partial_{i_n}^{\beta_n}$ and β ranges over multi-indices on the active variables.

Proof. Set $v(z) = \sum_{j=0}^n z_j e_{i_j}$. The symmetric p-linearity of $D^p u(x)$ and the multinomial theorem give

$$\begin{aligned} D^p u(x)\big[v(z), \dots, v(z)\big] &= \sum_{|\beta|=p} \binom{p}{\beta} z^\beta\, \partial^\beta u(x) \\ &= \sum_{|\beta|=p} \frac{p!}{\beta!} z^\beta\, \partial^\beta u(x). \end{aligned}$$

By definitions (4) and (5), $Q_u(z) = D^p u(x)[v(z), \dots, v(z)] \,/\, p!$. Dividing the displayed multinomial expansion by $p!$ proves (6). ☐

Corollary 2.2. *$\partial^\nu u(x) = \nu!\, [z^a]\, Q_u(z)$, where $a = (a_0, \dots, a_n)$ and $[z^a]$ extracts the coefficient of $z^a = z_0^{a_0} \cdots z_n^{a_n}$.*

Proof. Read off the coefficient of z^a in (6) and use $a! = \nu!$; in active coordinates $\partial^a u(x) = \partial^\nu u(x)$. ☐

Extracting one mixed derivative is therefore a linear functional on $\mathbb{R}[z]_p$, and the remaining question is how few directional evaluations suffice to realize it.

2.2. Directional schedules

A *directional schedule* of length R for ∂^ν is a finite collection $(c_r, v_r)_{r=1}^R \subset \mathbb{R} \times V$ such that

$$\partial^\nu u(x) \;=\; \sum_{r=1}^{R} c_r\, T_p(x; v_r) \tag{7}$$

for every $u \in C^p$ on a neighborhood of x.

Because the schedule is required to hold for every such u, the p-jet of u at x may be prescribed arbitrarily on the active variables: for any coefficients q_β, the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(x) = \beta!\, q_\beta$. This observation lets us test identities coefficient by coefficient in $\mathbb{R}[z]_p$.

Two questions now organize the rest of the paper. The analytic one, which we label problem (P), asks for the minimum length R^* of a directional schedule (7) for ∂^ν and for an explicit schedule of that length. The algebraic one, problem (Q), asks for the smallest number of pth powers of linear forms whose linear combination equals $p!\, z^a$. Section 3 shows that (P) and (Q) are the same problem, so that the classical answer to (Q) settles (P).

3. Equivalence to monomial Waring decomposition

3.1. The bridge theorem

For $v \in V$, write $v = \sum_{j=0}^n v_j e_{i_j}$ in active coordinates. We pair this direction with the polynomial variable $z = (z_0, \ldots, z_n)$ by $v \cdot z = \sum_{j=0}^n v_j z_j$.

Theorem 3.1 (Bridge)**.** *Let $R \in \mathbb{N}$, $c_r \in \mathbb{R}$, $v_r \in V$, $r = 1, \ldots, R$. The following are equivalent.*

(1) *(Directional schedule)* $\partial^\nu u(x) = \sum_{r=1}^R c_r T_p(x; v_r)$ *for every $u \in C^p$ near x.*

(2) *(Polynomial identity in $\mathbb{R}[z]_p$)* $p!\, z^a = \sum_{r=1}^R c_r (v_r \cdot z)^p$.

Proof. Write $v_r = (v_{r,0}, \ldots, v_{r,n})$ in active coordinates. Lemma 2.1 gives

$$T_p(x; v_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!} v_r^\beta.$$

Therefore, the right-hand side of the directional-schedule identity is

$$\sum_{r=1}^{R} c_r T_p(x; v_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!} \left(\sum_{r=1}^{R} c_r v_r^\beta \right).$$

The order-p derivatives $(\partial^\beta u(x))_{|\beta|=p}$ can be prescribed independently by choosing a polynomial with the desired p-jet at x. Since the target is $\partial^\nu u(x) = \partial^a u(x)$ in active coordinates, the directional-schedule identity holds for every u if and only if

$$\sum_{r=1}^{R} c_r v_r^\beta = \nu!\,\delta_{a\beta}, \quad |\beta| = p. \tag{$\star$}$$

It remains to show that the polynomial identity gives the same coefficient conditions. Expanding its right-hand side yields

$$\sum_{r=1}^{R} c_r (v_r \cdot z)^p = \sum_{|\beta|=p} \binom{p}{\beta} \left(\sum_{r=1}^{R} c_r v_r^\beta \right) z^\beta.$$

Hence $p!\,z^a = \sum_{r=1}^{R} c_r (v_r \cdot z)^p$ holds if and only if

$$\binom{p}{\beta} \sum_{r=1}^{R} c_r v_r^\beta = p!\,\delta_{a\beta}, \quad |\beta| = p.$$

Dividing by $\binom{p}{\beta}$ gives $\sum_{r=1}^{R} c_r v_r^\beta = \beta!\,\delta_{a\beta}$. Because $\beta!\,\delta_{a\beta} = \nu!\,\delta_{a\beta}$, this is exactly ($\star$). Thus the directional-schedule identity and the polynomial identity are equivalent. $\square$

Definition 3.2 (Waring rank)**.** For a homogeneous polynomial $F \in \mathbb{R}[z]_p$, its *real Waring rank* is

$$R_{\mathbb{R}}(F) \;:=\; \min \Big\{ R : \; F = \sum_{r=1}^{R} c_r L_r(z)^p, \; c_r \in \mathbb{R}, \; L_r \in \mathbb{R}[z]_1 \Big\}.$$

The *complex Waring rank* $R_{\mathbb{C}}(F)$ is defined identically with $\mathbb{R}$ replaced by $\mathbb{C}$; since $\mathbb{R} \subset \mathbb{C}$, $R_{\mathbb{C}}(F) \le R_{\mathbb{R}}(F)$.

Corollary 3.3. *Among schedules with real coefficients and real directions, the minimum length R^* for ∂^ν equals the real Waring rank $R_{\mathbb{R}}(z^a)$ of the monomial.*

Proof. By Theorem 3.1, schedules of length R for ∂^ν correspond to decompositions of $p!\,z^a$ into R pure pth powers, and absorbing the constant $p!$ into the coefficients c_r shows that $R_{\mathbb{R}}(p!\,z^a) = R_{\mathbb{R}}(z^a)$. $\square$

Remark 3.4 (Complex extension)**.** Using the complexified directional coefficient defined after (4), the same coefficient comparison holds for $c_r \in \mathbb{C}$, $v_r \in V_{\mathbb{C}}$, and identities in $\mathbb{C}[z]_p$. The minimum length of a complex schedule is therefore $R_{\mathbb{C}}(z^a)$.

3.2. The Waring rank of a monomial

Problems (P) and (Q) were posed over $\mathbb{R}$, and Remark 3.4 admits complex coefficients and directions as well. Enlarging the field can only decrease the rank, so $R_{\mathbb{C}} \le R_{\mathbb{R}}$, and the smaller of the two ranks has a classical closed form.

Theorem 3.5 (Carlini–Catalisano–Geramita [8])**.** *For $z^a = z_0^{a_0} \cdots z_n^{a_n}$ with $1 \le a_0 \le \cdots \le a_n$,*

$$R_{\mathbb{C}}(z^a) \;=\; \prod_{j=1}^{n}(a_j+1) \;=\; \frac{\prod_{j=0}^{n}(a_j+1)}{a_0+1}. \tag{8}$$

The lower bound uses the apolar ideal $(z^a)^{\perp} = (\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1})$, which is a complete intersection; the apolar interpretation is recalled in Appendix A.3. The upper bound is constructive, and Theorem 3.6 states the construction we implement.

For $m \ge 1$, write $U_m := \{\zeta \in \mathbb{C} : \zeta^m = 1\}$.

Theorem 3.6 (Roots-of-unity directional schedule)**.** *Let $m_j = a_j + 1$ for $j = 1, \ldots, n$, and for each tuple $\zeta = (\zeta_1, \ldots, \zeta_n) \in U_{m_1} \times \cdots \times U_{m_n}$ define*

$$v_\zeta = e_{i_0} + \sum_{j=1}^{n} \zeta_j e_{i_j}, \qquad c_\zeta = \frac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j. \tag{9}$$

Then $\partial^\nu u(x) = \sum_\zeta c_\zeta\, T_p(x; v_\zeta)$ for every $u \in C^p$ near x. The schedule has length $\prod_{j=1}^{n} m_j = R_{\mathbb{C}}(z^a)$ and is therefore length-optimal over $\mathbb{C}$.

The proof, given in Appendix A.2, verifies condition (2) of Theorem 3.1 directly via the roots-of-unity filter $\sum_{\zeta^m=1} \zeta^k = m\, \mathbf{1}_{m|k}$.

To illustrate how Theorem 3.1 couples problem (P), the analytic identity (7), with problem (Q), the polynomial identity $p!\, z^a = \sum_r c_r (v_r \cdot z)^p$, we work out two examples: a low-order one for which both sides reduce to hand-checkable identities, and a higher-order repeated-index one that exhibits the full strength of the complex Waring construction.

Example 3.7 ($p = 2$, $a = (1,1)$, real schedule of length 2)**.** For $a = (1,1)$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, $U_2 = \{+1, -1\}$. Equation (9) produces the schedule

$$\bigl(c_{+1}, v_{+1}\bigr) = \bigl(\tfrac{1}{2},\, e_1 + e_2\bigr), \qquad \bigl(c_{-1}, v_{-1}\bigr) = \bigl(-\tfrac{1}{2},\, e_1 - e_2\bigr). \tag{10}$$

Verification on the analytic side. The directional Taylor coefficient is $T_p(x;v) = \frac{1}{2}\big(u_{11}v_1^2 + 2u_{12}v_1v_2 + u_{22}v_2^2\big)$, so the two probes give $T_p(x; e_1 \pm e_2) = \frac{1}{2}\big(u_{11} \pm 2u_{12} + u_{22}\big)$ and the diagonal terms cancel in the difference:

$$\tfrac{1}{2}T_p(x; e_1 + e_2) - \tfrac{1}{2}T_p(x; e_1 - e_2) = u_{12}(x).$$

Verification on the algebraic side. The equivalent polynomial identity in $\mathbb{R}[z]_2$ reads

$$\begin{aligned} 2!\, z_0 z_1 &= \tfrac{1}{2}(z_0 + z_1)^2 - \tfrac{1}{2}(z_0 - z_1)^2 \\ &= \tfrac{1}{2}\big[(z_0^2 + 2z_0z_1 + z_1^2) - (z_0^2 - 2z_0z_1 + z_1^2)\big] \\ &= 2\, z_0 z_1, \end{aligned}$$

matching $p!\, z^a = 2\, z_0 z_1$. The two checks coincide via the bridge of Theorem 3.1.

Example 3.8 ($p = 6$, $a = (2,2,2)$, complex schedule of length 9)**.** Consider $u_{112233} = \partial_1^2 \partial_2^2 \partial_3^2 u$, i.e. $a = (2,2,2)$. Here $n = 2$, $a_0 = a_1 = a_2 = 2$, $m_1 = m_2 = 3$; take $\omega = e^{2\pi \mathrm{i}/3}$ and $U_3 = \{1, \omega, \omega^2\}$. Equation (9) produces $R_{\mathbb{C}}(z^{(2,2,2)}) = 9$ directions: for each $(\zeta_1, \zeta_2) \in U_3 \times U_3$,

$$v_\zeta \;=\; e_1 + \zeta_1 e_2 + \zeta_2 e_3, \qquad c_\zeta \;=\; \frac{(2!)^3}{3 \cdot 3}\,\zeta_1\zeta_2 \;=\; \frac{8}{9}\,\zeta_1\zeta_2. \tag{11}$$

Analytic side. The directional schedule reads

$$u_{112233}(x) \;=\; \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1\zeta_2\, T_p\big(x;\, e_1 + \zeta_1 e_2 + \zeta_2 e_3\big). \tag{12}$$

Algebraic side. The equivalent polynomial identity in $\mathbb{C}[z]_6$ is

$$6!\, z_0^2 z_1^2 z_2^2 \;=\; \frac{8}{9} \sum_{\zeta_1, \zeta_2 \in U_3} \zeta_1\zeta_2\, (z_0 + \zeta_1 z_1 + \zeta_2 z_2)^6. \tag{13}$$

That (13) holds is verified by the roots-of-unity filter (Lemma A.1): expanding the right-hand side multinomially and collecting the coefficient of $z_0^{\beta_0} z_1^{\beta_1} z_2^{\beta_2}$ reduces to the product $\prod_{j=1}^2 \sum_{\zeta_j \in U_3} \zeta_j^{1+\beta_j}$, which is nonzero exactly when $\beta_j \equiv 2 \pmod 3$ for $j = 1, 2$; together with $\sum_j \beta_j = 6$ this forces $\beta = (2,2,2)$, and the leading constant $\frac{8}{9} \cdot \frac{6!}{2!\,2!\,2!} \cdot 9 = 720 = 6!$ gives the right normalization.

By comparison, the coalesced antipodally merged real polarization schedule used in our baseline requires 13 distinct nonzero directions for the same $u_{112233}(x)$; the complex Waring schedule above attains the complex Waring-rank lower bound $R_{\mathbb{C}} = 9$ (Theorem 3.5).

3.3. Pattern taxonomy

Table 1 lists direction counts for representative active-exponent patterns up to $p = 6$. The complex rank $R_{\mathbb{C}}$ is Eq. (8), and the two real counts come from Eq. (18) of Section 4.2 under two merging rules: backend B2, the rule we implement, merges duplicate and antipodal integer directions, whereas projective consolidation additionally uses $T_p(x; \lambda v) = \lambda^p T_p(x; v)$ to merge all proportional directions. In the table the complex count never exceeds a real one, and it falls strictly below R_{proj} exactly for the three patterns in which at least two coordinates carry a repeated exponent.

Table 1: Directional counts for representative active-exponent patterns. B2 is the implemented duplicate/antipodal coalescer; R_{proj} additionally merges proportional directions. Bold complex ranks are strictly smaller than R_{proj}. The square-free pattern (1^p) receives no reduction.

pattern	example multi-index	$R_{\mathbb{C}}$	R_{B2}	R_{proj}
(p)	$\partial_1^p u$	1	$\lceil p/2 \rceil$	1
$(2,1)$	u_{112}	3	3	3
$(1,1,1)$	u_{123}	4	4	4
$(3,1)$	u_{1112}	4	4	4
$(2,2)$	u_{1122}	**3**	4	4
$(2,1,1)$	u_{1123}	6	6	6
$(1,1,1,1)$	u_{1234}	8	8	8
$(4,2)$	u_{111122}	**5**	7	6
$(2,2,2)$	u_{112233}	**9**	13	13
(1^6)	u_{123456}	32	32	32

4. Taylor-mode evaluation and backend dispatch

4.1. The Taylor jet for Linear–sinh MLPs

Given a single direction v, Theorem 3.6 reduces the problem to evaluating $T_p(x; v)$. We use forward-mode Taylor propagation [2, 3], replacing each intermediate activation $y \in \mathbb{K}^B$ over a batch of B points, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, by a length-$(p+1)$ tuple $(y_0, \ldots, y_p)$ representing its truncated Taylor coefficients $y_k = (1/k!)\, d^k y/d\tau^k|_{\tau=0}$.

For an MLP $u = \phi_L \circ A_L \circ \cdots \circ \phi_1 \circ A_1$ with affine A_ℓ and entrywise activations ϕ_ℓ, the jet rules are as follows. The affine layer $y = Wx + b$

propagates each jet order independently:

$$y_0 \;=\; W x_0 + b, \qquad y_k \;=\; W x_k \quad (k \geq 1). \tag{14}$$

Each nonlinearity is propagated together with one auxiliary jet that closes its derivative recurrence. For tanh the auxiliary jet is $z = 1 - y^2$, initialized by $y_0 = \tanh(x_0)$ and $z_0 = 1 - y_0^2$, and

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; -\sum_{j=0}^{k} y_j\, y_{k-j}, \qquad k \geq 1. \tag{15}$$

For sinh the auxiliary jet is $z = \cosh(x)$, initialized by $y_0 = \sinh(x_0)$ and $z_0 = \cosh(x_0)$, and the coupled sinh / cosh recurrence is

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, y_j\, x_{k-j}, \qquad k \geq 1. \tag{16}$$

The sinh / cosh recurrence has the same coupled form as the sin / cos recurrence, but the cosh update has no sign flip. These recurrences are algebraic identities and can be evaluated on complex tensors. We use sinh for complex-parameter networks because sinh and cosh are entire. The implementation also contains sin and sigmoid recurrences for the benchmark architectures. Although the local jet of tanh is well defined away from a pole, tanh is meromorphic rather than entire.

Each Taylor coefficient y_k is a regular tensor, and the jet propagation does not nest input-gradient graphs. It stores $p + 1$ coefficient tensors per activation. We additionally use a custom `torch.autograd.Function` for each activation jet, with the closed-form backward

$$g_x^{(m)} \;=\; \sum_{j=0}^{p-m} \overline{q_j}\, g_y^{(m+j)}, \qquad q(\tau) \;=\; \phi'(x(\tau)), \tag{17}$$

where q_j is the jth Taylor coefficient of q. This vector–Jacobian product avoids reverse differentiation through the intra-jet recurrence. Equation (17) uses PyTorch's complex-gradient convention for a real-valued loss; the conjugation is the Hermitian adjoint of elementwise multiplication by q_j.

4.2. Backends

We compare three implemented deterministic backends for evaluating one fixed $\partial^\nu u(x)$. All three are differentiable in the network parameters θ, so each can be used directly as the residual term of a PINN loss. The implementation also provides an `auto` dispatch option, but it is a selection rule between B2 and B3 rather than a separate mathematical backend.

(B1) Direct nested AD. The reference baseline performs p successive calls to PyTorch's `autograd.grad`, each with retained computation graph. This induces a reverse-mode autograd graph of depth $O(pL)$ for a depth-L network.

(B2) Real polarization with Taylor jet. The classical antipodally merged polarization identity is

$$\partial^\nu u(x) \;=\; \frac{1}{2^p} \sum_{\epsilon\in\{\pm1\}^p} \Big(\textstyle\prod_{r=1}^{p} \epsilon_r \Big)\, T_p\Big(x;\, \textstyle\sum_{r=1}^{p} \epsilon_r\, e_{i_r}\Big). \tag{18}$$

After merging duplicate and antipodal integer directions, B2 yields at most 2^{p-1} real directions. It does not apply the additional projective normalization reported in Table 1. The directional Taylor coefficients $T_p(x;\,\cdot\,)$ are evaluated by the jet propagation of Section 4.

(B3) Complex Waring with Taylor jet. The roots-of-unity construction of Theorem 3.6 produces $R_{\mathbb{C}}(z^a) = \prod_{j=1}^{n}(a_j + 1)$ directions in $\mathbb{C}^{n+1}$; for several repeated-index patterns this is smaller than the coalesced polarization count of (B2). For real-parameter networks, the complex directions induce differentiable casts of the linear weights via `Tensor.to`; for complex-parameter networks no cast is required and the network itself acts on the complex directions natively.

4.3. The end-to-end algorithm

Algorithm 1 summarizes the complete pipeline of backend (B3): given a scalar MLP whose primitives admit the implemented complex jets, a point x, and a multi-index ν of order p, it returns the value $\partial^\nu u(x)$ without nested input differentiation. The implementation batches all schedule directions in one jet pass. Backend (B2) follows the same overall structure with the Waring schedule of lines 3–8 replaced by the polarization schedule (18).

The schedule tensors depend on ν, the device, and the scalar type but not on the trainable parameters. The reference implementation constructs them

Algorithm 1 Single-monomial $\partial^\nu u(x)$ using a complex Waring schedule and a batched Taylor jet.

Require: scalar MLP u built from supported analytic primitives; point $x \in \mathbb{R}^d$; multi-index ν of order p

Ensure: the value $\partial^\nu u(x)$

relabel the active coordinates of ν as $i_0, \dots, i_n$ with exponents $1 \le a_0 \le \cdots \le a_n$
$m_j \leftarrow a_j + 1$ for $j = 1, \dots, n$; set $\mathcal{I} \leftarrow U_{m_1} \times \cdots \times U_{m_n}$
initialize schedule $\mathcal{S} \leftarrow \emptyset$
for all $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathcal{I}$ **do**
    $v_\zeta \leftarrow e_{i_0} + \sum_{j=1}^{n} \zeta_j \, e_{i_j}$
    $c_\zeta \leftarrow \dfrac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j$
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{(c_\zeta, v_\zeta)\}$
end for
stack the directions into $V_{\mathcal{S}} \in \mathbb{C}^{|\mathcal{S}| \times d}$
form the batched input jet $\mathbf{J}_0 \leftarrow (x \text{ replicated}, V_{\mathcal{S}}, 0, \dots, 0)$
$\mathbf{J}_L \leftarrow$ propagate $\mathbf{J}_0$ through u using (14)–(16)
$t \leftarrow [\mathbf{J}_L]_p$ ▷ one order-p coefficient per direction
return $\sum_{r=1}^{|\mathcal{S}|} c_r t_r$

at call entry and includes that work in measured throughput; a caller may cache a fixed schedule instead. For a real model, conjugate pairing makes the returned value real in exact arithmetic; the implementation nevertheless retains the complex tensor and takes its real part only when the target PDE is real-valued.

Remark 4.1. By Theorem 3.1 together with Theorem 3.6, Algorithm 1 returns $\partial^\nu u(x)$ algebraically exactly; floating-point evaluation introduces round-off. The batch contains $|\mathcal{S}| = R_{\mathbb{C}}(z^a)$ directions, the minimum possible for a complex pure-power formula (Theorem 3.5).

4.4. Cost and numerical considerations

For a fixed direction, the Taylor jet stores $p+1$ coefficient tensors per activation and applies each linear map to the stacked coefficients; there is no chain of nested reverse-mode differentiation through the input coordinates. Affine propagation is linear in p, while the activation recurrences require

$\sum_{k=1}^{p} O(k) = O(p^2)$ elementwise products per scalar and direction. The total work and directional storage scale linearly with $R_{\mathbb{C}}(z^a)$ for B3 and with the implemented coalesced count R_{B2} for B2. The rank reduction is an algebraic cost reduction; the realized wall-clock time also depends on complex arithmetic, batching, activation kernels, and whether parameter gradients are required.

The schedule coefficients are roots of unity multiplied by $\nu!/R_{\mathbb{C}}$, so coefficient magnitudes are explicit and do not come from solving an ill-conditioned linear system. The method has no finite-difference step size and no Monte Carlo variance. Remaining errors are ordinary floating-point round-off from Taylor recurrences, complex arithmetic, and the final summation over directions. Avoiding a linear solve does not eliminate cancellation, factorial scaling, or sinh overflow; the present verification covers double precision through order six and is not a general conditioning study.

5. Implementation verification

We verify the algebraic schedule, the derivative values, and the parameter gradients separately. All checks use double precision, and all are executable tests in the repository. First, the roots-of-unity coefficient filter is evaluated against every degree-four monomial in five variables for the active-exponent pattern $(2, 1, 1)$; the maximum admissible absolute coefficient error is 10^{-12}. The direction counts for all patterns in Table 1 are also checked against the closed-form rank and the B2 coalescing implementation.

Second, B2 and B3 are compared with B1 on real-parameter networks using tanh, sigmoid, sin, and sinh activations. The tests cover the patterns $(2, 2)$, $(1, 1, 1)$, and (3). Values must agree with nested automatic differentiation to relative tolerance 2×10^{-10} and absolute tolerance 2×10^{-11}; the imaginary remainder for a real target is bounded by 2×10^{-11}.

Third, gradients of a squared derivative loss with respect to every network parameter are compared between the jet backends and nested differentiation. The tolerance is 2×10^{-9} relative and 2×10^{-11} absolute for both real- and complex-parameter networks. Additional analytic checks cover orders $1, 2, 3, 4$, and 6 for a scalar complex sinh network. Every check passes at the tolerance stated for it.

These tests establish implementation consistency, but they do not convert a direction-count reduction into a hardware-independent timing claim. The application study below therefore reports measured throughput together with accuracy.

6. Numerical experiments on high-order and complex-valued PDEs

This section evaluates the Taylor-jet implementation in complete PINN workflows. The objective is not to time an isolated derivative call, but to measure the accuracy reached under a fixed compute budget when the derivative backend is embedded in training. We compare a complex-parameter sinh network with SIREN [9], a multiscale Fourier-feature PINN (mFF-PINN) [10, 11], and MscaleDNN-2-sin [12]. The three baseline implementations were traced to pinned upstream commits and tested for architecture and derivative fidelity.

6.1. Shared protocol

We call one problem together with one value of its sweep parameter a *setting*; the protocol below comprises 12 settings. Protocol `jsc_v2` grants a wall-clock budget of 1200 s to each triple of setting, method, and seed, and uses five seeds, four trainable hidden layers, 4096 fixed interior points, 512 fixed boundary points, paired per-seed collocation, and a common held-out set of 8192 points. Every hidden layer has width $H = 128$. This shared-width control does not match trainable parameter counts: each complex parameter counts as two real degrees of freedom, frozen Fourier maps are excluded, and a split-real Maxwell model contains two width-128 networks. The resulting counts are reported with each table.

All methods use Adam with initial learning rate 10^{-3}, cosine decay to 10^{-4}, and the same residual normalization and boundary penalty within a setting. The four methods receive the same collocation points for a given seed. Complex sinh uses complex128 parameters; real targets are fitted through its real output component, with a 10^{-6} penalty on imaginary parameters. For Maxwell, the full complex output is fitted and each real baseline represents real and imaginary components with separate networks.

Architecture-specific frequencies. For polyharmonic problems, the Complex sinh first-layer scale is 2π, while the mFF-PINN branches use scales 1 and π. For chirp and Maxwell problems, these become $\max(10, 2\pi a)$ and $(1, \max(2, \pi a))$, respectively. SIREN retains its upstream first- and hidden-layer frequencies of 30. MscaleDNN-2-sin uses three independent subnets at fixed input scales $1, 2$, and 4.

Derivative backend and metric. All four architectures use B3 for every monomial derivative. Scaled activations and branched embeddings are admitted only after their input derivatives and parameter gradients agree with direct automatic differentiation. Accuracy is the held-out relative L^2 error

$$\mathcal{E}_{L^2} = \frac{\|u_\theta - u_\star\|_{L^2(\Omega)}}{\|u_\star\|_{L^2(\Omega)}}. \tag{19}$$

Tables report the arithmetic mean and sample standard deviation over five seeds; history panels show the geometric mean and seedwise range. A setting is reported only after automated checks confirm all 20 method–seed rows, that is, four methods and five seeds, together with finite metrics, monotone histories, width and protocol metadata, and a fixed evaluation rule. All runs used an NVIDIA H20 GPU with CUDA 12.1; the stored records specify PyTorch 2.4.1 or 2.5.1, so throughput is compared within a setting rather than across settings.

6.2. Polyharmonic benchmark

For $d \in \{2, 3\}$, let

$$u_\star(x) = \prod_{j=1}^{d} \sin(\pi x_j), \qquad \Delta^m u_\star = (-d\pi^2)^m u_\star \quad \text{in } (-1,1)^d. \tag{20}$$

We use orders $2m \in \{2, 4, 6\}$ and enforce the Navier targets $u = u_\star$ and $\Delta^j u = (-d\pi^2)^j u_\star$, $j = 1, \ldots, m-1$, on the boundary; all these targets vanish for the chosen $u_\star$. Expanding Δ^m produces $\binom{d+m-1}{m}$ monomial terms. Each is evaluated independently; mixed terms such as $\partial_x^2 \partial_y^2 u$ use the Waring schedule, while pure coordinate powers need one complex direction.

Table 2: Polyharmonic benchmark, $d = 2$: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.36 \pm 1.57} \times 10^{-4}$	1.00 ± 0.00	$4.88 \pm 2.73 \times 10^{-3}$	$1.18 \pm 0.24 \times 10^{-2}$
4	$\mathbf{8.02 \pm 1.86} \times 10^{-3}$	1.00 ± 0.00	$8.18 \pm 1.29 \times 10^{-1}$	$5.74 \pm 0.45 \times 10^{-1}$
6	$\mathbf{1.01 \pm 0.53} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.02	$9.39 \pm 0.11 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

Table 3: Polyharmonic benchmark, $d = 3$: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.29 \pm 0.39} \times 10^{-3}$	1.00 ± 0.00	$4.61 \pm 1.39 \times 10^{-1}$	$2.33 \pm 0.25 \times 10^{-1}$
4	$\mathbf{1.57 \pm 0.73} \times 10^{-1}$	1.00 ± 0.00	$9.95 \pm 0.04 \times 10^{-1}$	$9.38 \pm 0.35 \times 10^{-1}$
6	$\mathbf{9.82 \pm 0.09} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.01	1.07 ± 0.07

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,354; SIREN=50,177; mFF-PINN=66,305; MscaleDNN-2-sin=150,531.

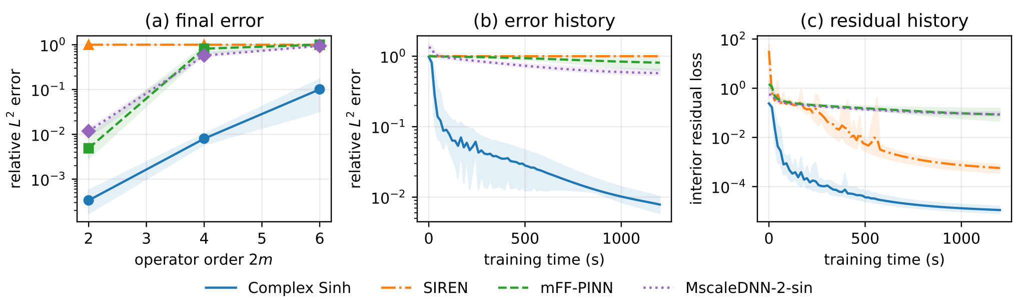

Figure 1: Two-dimensional polyharmonic benchmark. Panel (a) shows the five-seed mean relative L^2 error and seedwise range after 1200 s. Panels (b) and (c) show error and normalized interior residual histories for the representative fourth-order setting.

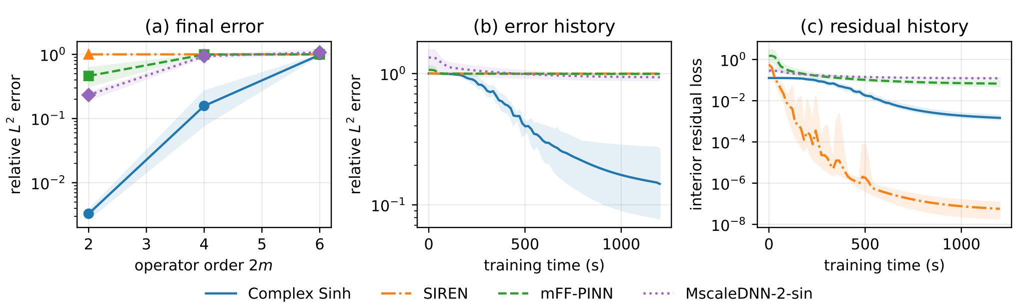

Figure 2: Three-dimensional polyharmonic benchmark, with the same convention as Fig. 1. History panels use the fourth-order setting.

In two dimensions, the mean error of Complex sinh increases from 3.36×10^{-4} at order two to 1.01×10^{-1} at order six, but remains below the three real baselines throughout. The gap is largest at order four, where its mean error is approximately 72 times smaller than the next lowest mean. In three dimensions the same ordering holds at orders two and four. At order six, however, Complex sinh reaches 9.82×10^{-1}, SIREN reaches approximately 1, and the other baselines are also near unit error. This setting demonstrates budget-limited saturation rather than a practically meaningful separation.

6.3. Non-separable radial chirp

On $(-1,1)^2$, define

$$u_\star(x,y) = \sin\left(\frac{a\pi}{2}(x^2+y^2)\right), \qquad -\Delta u + u = f, \tag{21}$$

with Dirichlet data from $u_\star$ and $a \in \{1,2,3\}$. Writing $\phi = a\pi(x^2+y^2)/2$ and $r^2 = x^2+y^2$, the manufactured source is

$$f = (a\pi)^2 r^2 \sin\phi - 2a\pi\cos\phi + \sin\phi. \tag{22}$$

Unlike a separable Fourier mode, this target has local radial wavenumber $|\nabla\phi| = a\pi r$, so its frequency varies over the domain.

Table 4: Radial-chirp benchmark: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
1	$\mathbf{8.14 \pm 7.00} \times 10^{-4}$	$4.25 \pm 0.06 \times 10^{-1}$	$2.56 \pm 1.84 \times 10^{-3}$	$1.24 \pm 0.24 \times 10^{-2}$
2	$\mathbf{2.39 \pm 0.57} \times 10^{-3}$	1.53 ± 0.02	$9.69 \pm 1.00 \times 10^{-1}$	$8.47 \pm 1.85 \times 10^{-1}$
3	$\mathbf{1.00 \pm 0.18} \times 10^{-1}$	1.07 ± 0.02	1.42 ± 0.07	$9.02 \pm 0.97 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

Complex sinh gives the lowest mean error at each chirp rate. At $a = 1$, its mean error of 8.14×10^{-4} is about three times smaller than that of mFF-PINN, the next most accurate method. The separation grows at $a = 2$, where the three real baselines have mean errors between 0.847 and 1.53, while Complex sinh remains at 2.39×10^{-3}. At $a = 3$, its error rises to 0.100 but stays below all three real baselines. Because all methods share one loss and optimizer protocol, these results compare end-to-end methods; they do not by themselves identify representation as the sole cause of the gap.

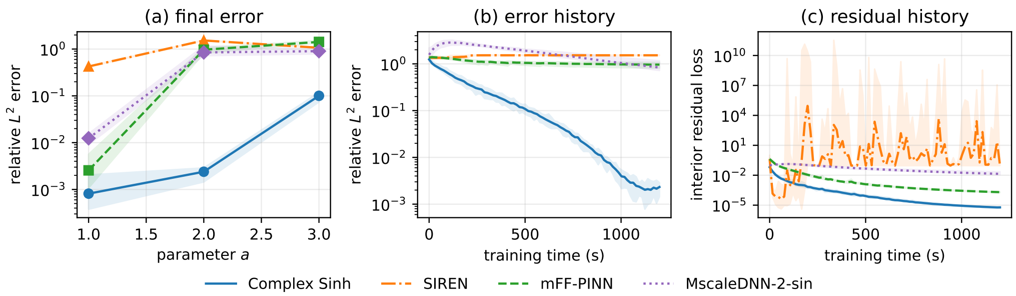

Figure 3: Radial-chirp benchmark. Panel (a) reports final five-seed accuracy and seedwise ranges. Panels (b) and (c) show accuracy and normalized residual histories for $a = 2$.

6.4. Time-harmonic Maxwell benchmark

For a two-dimensional transverse-magnetic (TM_z) mode, Maxwell's equations reduce to a scalar complex Helmholtz equation for the out-of-plane field E:

$$\Delta E + \kappa^2 E = f, \qquad \kappa^2 = (a\pi)^2(1 + \mathrm{i}\beta), \qquad E_\star = \exp\bigl(\mathrm{i}a\pi(x + y)\bigr), \quad (23)$$

with $\beta = 0.2$, exact Dirichlet data, and $a \in \{2, 4, 6\}$. The source is

$$f = \left[-2(a\pi)^2 + \kappa^2\right] E_\star.$$

Complex sinh represents E natively; each real baseline uses a split real–imaginary pair of two $H = 128$ networks.

Table 5: Time-harmonic Maxwell benchmark: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{4.21 \pm 1.57} \times 10^{-3}$	1.00 ± 0.00	$6.93 \pm 1.86 \times 10^{-1}$	1.09 ± 0.06
4	$\mathbf{4.19 \pm 0.22} \times 10^{-2}$	1.00 ± 0.00	1.06 ± 0.11	1.29 ± 0.07
6	$\mathbf{1.20 \pm 0.17} \times 10^{-1}$	1.00 ± 0.00	1.13 ± 0.15	1.34 ± 0.04

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=100,098; mFF-PINN=132,610; MscaleDNN-2-sin=300,294.

The native-complex network has the lowest mean error at every wavenumber. Its mean error increases from 4.21×10^{-3} at $a = 2$ to 0.120 at $a = 6$. The split-real baselines remain near or above unit error at the two higher

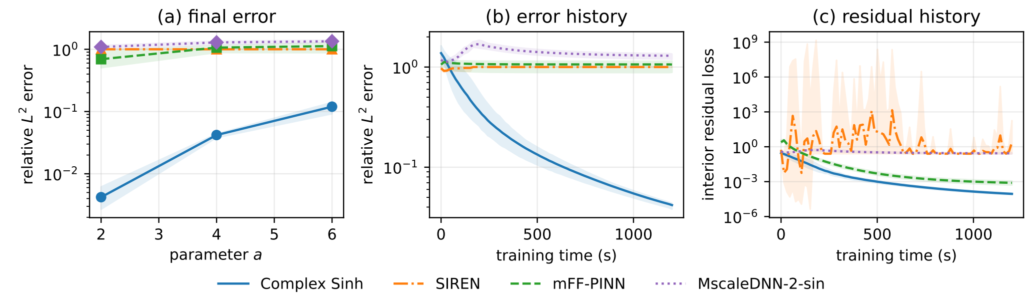

Figure 4: Time-harmonic Maxwell benchmark. Panel (a) reports final five-seed accuracy and seedwise ranges. Panels (b) and (c) show accuracy and normalized residual histories for $a = 4$.

wavenumbers. This experiment favors a native complex representation structurally, but the comparison still includes architecture-dependent optimization and throughput effects; it should not be interpreted as a controlled ablation of complex arithmetic alone.

6.5. Throughput and aggregate outcome

Table 6: Representative optimizer throughput: mean $\pm$ sample standard deviation of milliseconds per step over five seeds. Methods should be compared within a row; the stored software versions differ between settings.

setting	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
Polyharmonic, $d = 2$, $2m = 4$	243.1 ± 0.2	243.7 ± 0.2	576.8 ± 0.2	729.7 ± 0.4
Radial chirp, $a = 2$	70.7 ± 0.2	61.9 ± 0.0	145.2 ± 0.1	182.3 ± 0.0
Maxwell, $a = 4$	70.8 ± 0.1	123.5 ± 0.0	290.1 ± 0.0	367.8 ± 0.1

The representative throughput rows show why equal wall-clock time is a more informative control than equal optimizer steps. On the scalar chirp task, SIREN completes the fastest steps, whereas the branched mFF-PINN and the three MscaleDNN subnets are slower. For Maxwell, each split-real baseline evaluates two networks, increasing its per-step cost. The polyharmonic row also shows the steep cost of propagating fourth-order jets through branched architectures.

Across the 12 settings, Complex sinh attains the smallest five-seed mean error in every table. In 11 of them, the next-lowest mean is at least three times larger. The exception is the three-dimensional sixth-order polyharmonic setting, where all methods are close to the trivial-predictor error. This distinction is retained in the discussion rather than reducing the benchmark to an unqualified win count.

7. Discussion and limitations

7.1. Direction count and computational cost

Because the rank formula (8) divides out the factor a_0+1 belonging to the smallest active exponent, the complex schedule gains most when every active coordinate is repeated, that is, when $a_0 \geq 2$; examples are $(2,2)$, $(2,2,2)$, $(3,3)$, and $(4,2)$. For $(2,2,2)$ it uses 9 directions rather than the 13 produced by coalesced real polarization. This reduction controls the batch size of the directional jet, but not every component of execution time. Complex arithmetic, kernel launch overhead, memory traffic, and the backward pass can dominate for small networks or small batches. The rank theorem should therefore be read as an exact algebraic complexity result, not as a hardware-independent speedup.

7.2. Interpretation of the PINN study

Under the frozen `jsc_v2` protocol, Complex sinh has the lowest mean error in all 12 benchmark settings. The margin to the next lowest mean exceeds a factor of 70 in four of them: the two-dimensional fourth-order and three-dimensional second-order polyharmonic problems, the radial chirp at $a = 2$, and Maxwell at $a = 2$. The three-dimensional sixth-order problem is a necessary counterweight: all methods remain near unit relative error, and the difference between Complex sinh and SIREN is practically negligible.

The comparison does not isolate one causal mechanism. A complex model changes the value representation, initialization, parameter count, and throughput at the same time. Literal width $H = 128$ gives the scalar Complex sinh network exactly twice as many real degrees of freedom as SIREN, whereas MscaleDNN-2-sin has more degrees of freedom but completes fewer steps. The loss weights and optimizer are shared rather than tuned separately for every architecture. The reported result is therefore a controlled comparison of complete methods under one compute protocol, not proof that complex parameters are intrinsically superior.

7.3. Operator scope and stochastic alternatives

For square-free patterns $R_{\mathbb{C}} = 2^{p-1}$ coincides with the coalesced real count, so the complex schedule saves no directions. Operator sums can be expanded and evaluated monomial by monomial, as in the polyharmonic experiment, but this procedure is not jointly rank-optimal for the full sum. Trace-collapsing approaches [6, 7] can be preferable for Laplacian-only workloads.

The stochastic Taylor derivative estimator (STDE) [5] solves a strictly broader problem—arbitrary contraction of the order-k derivative tensor—via stochastic sparse jets driven by integer partitions. For one fixed multi-index ν, our schedule is algebraically exact, whereas STDE is unbiased with variance controlled by its sampling budget. The approaches are complementary: stochastic estimation addresses large contractions, while the Waring schedule provides a minimal deterministic formula for one monomial derivative.

8. Conclusion

We have shown that the construction of a directional schedule for one mixed partial derivative $\partial^\nu u$ of order p is exactly the problem of a Waring decomposition of the monomial z^a. The classical Carlini–Catalisano–Geramita rank theorem and its roots-of-unity construction yield an explicit, rank-optimal directional schedule over $\mathbb{C}$. Combined with batched Taylor-mode propagation and a custom vector–Jacobian product, it gives an algebraically exact, parameter-differentiable backend for high-order neural surrogates.

The method is at its most useful for repeated-index multi-indices, where the complex Waring rank is materially smaller than the coalesced real polarization count. For square-free patterns it offers no rank advantage; for operator sums it is applied term by term without a global optimality claim. The same entire-on-$\mathbb{C}$ implementation supports complex-parameter sinh PINNs. In the validated five-seed application study, that complete method gives the lowest mean error on all 12 tested settings, but all methods saturate on the hardest three-dimensional sixth-order case. These results support the method as a useful deterministic component for repeated high-order derivatives and complex-valued PDE surrogates, while preserving the need for operator-specific methods in broader contractions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and data availability

The implementation, executable verification tests, experiment configurations, validated per-seed records, and table and figure generators are available in the public apolarity repository. Each formal result records the protocol identifier, source commit, hardware and software environment, random seed, architecture metadata, completed steps, and evaluation rule.

Appendix A. Proofs of directional schedule identities

Appendix A.1. Roots-of-unity filter

Lemma A.1. *For every positive integer m and every integer k,*

$$\sum_{\zeta \in U_m} \zeta^k = m\,\mathbf{1}_{\{m|k\}}.$$

Proof. Let $\omega = \exp(2\pi \mathrm{i}/m)$. Every element of U_m is ω^ℓ for a unique $\ell \in \{0, \ldots, m-1\}$, so

$$\sum_{\zeta \in U_m} \zeta^k = \sum_{\ell=0}^{m-1} (\omega^k)^\ell.$$

If $m \mid k$, then $\omega^k = 1$, and the sum equals m. If $m \nmid k$, then $\omega^k \neq 1$, and the finite geometric-series formula gives

$$\sum_{\ell=0}^{m-1} (\omega^k)^\ell = \frac{1-\omega^{km}}{1-\omega^k} = 0,$$

because $\omega^m = 1$. This proves the stated identity. $\square$

Appendix A.2. Proof of Theorem 3.6

Proof. We use the standard conventions that an empty product equals 1 and that a Cartesian product over an empty index set contains one empty tuple. These conventions include the case $n = 0$, for which the construction has one direction. By Remark 3.4 and Theorem 3.1, it is enough to prove the polynomial identity

$$\sum_{\zeta \in U_{m_1} \times \cdots \times U_{m_n}} c_\zeta (v_\zeta \cdot z)^p = p!\, z^a. \tag{A.1}$$

Fix $\beta \in \mathbb{Z}_{\geq 0}^{n+1}$ with $|\beta| = p$, and denote the coefficient of z^β on the left-hand side of (A.1) by C_β. Using the definitions of c_ζ and v_ζ in (9), followed by the multinomial theorem, gives

$$\begin{aligned} C_\beta &= \frac{\nu!}{\prod_{j=1}^n m_j} \binom{p}{\beta} \sum_{\zeta_1 \in U_{m_1}} \cdots \sum_{\zeta_n \in U_{m_n}} \prod_{j=1}^n \zeta_j^{\beta_j+1} \\ &= \frac{\nu!}{\prod_{j=1}^n m_j} \binom{p}{\beta} \prod_{j=1}^n \left(\sum_{\zeta_j \in U_{m_j}} \zeta_j^{\beta_j+1} \right). \end{aligned}$$

Lemma A.1 shows that the jth factor is nonzero if and only if $m_j \mid \beta_j + 1$. Since $m_j = a_j + 1$, this condition is equivalent to $\beta_j \equiv a_j \pmod{m_j}$. Thus, for each $j = 1, \ldots, n$, there is an integer $t_j \geq 0$ such that

$$\beta_j = a_j + t_j m_j.$$

The degree constraint $|\beta| = p = \sum_{j=0}^n a_j$ then gives

$$\beta_0 = a_0 - \sum_{j=1}^n t_j m_j.$$

If some t_j were positive, then $m_j = a_j + 1 \geq a_0 + 1$, and consequently $\beta_0 \leq a_0 - m_j < 0$, contradicting $\beta_0 \geq 0$. Hence every t_j is zero, so $\beta_j = a_j$ for $j = 1, \ldots, n$, and the degree constraint also gives $\beta_0 = a_0$. Therefore, $C_\beta = 0$ whenever $\beta \neq a$.

For $\beta = a$, every roots-of-unity sum equals m_j, and the product cancels the denominator:

$$C_a = \nu! \binom{p}{a} = \left(\prod_{j=0}^n a_j! \right) \frac{p!}{\prod_{j=0}^n a_j!} = p!.$$

The left-hand side of (A.1) therefore has coefficient $p!$ at z^a and coefficient zero at every other degree-p monomial. This proves (A.1), and Theorem 3.1 yields the stated directional schedule. Finally, the schedule contains

$$|U_{m_1} \times \cdots \times U_{m_n}| = \prod_{j=1}^n m_j = R_{\mathbb{C}}(z^a)$$

directions, where the last equality is Theorem 3.5. Hence the schedule is length-optimal over $\mathbb{C}$. $\square$

Appendix A.3. Apolar pairing alternative

The Waring rank of a homogeneous polynomial $F \in \mathbb{C}[z]_p$ admits the apolar characterization

$$R_{\mathbb{C}}(F) \;=\; \min\bigl\{\deg X :\; X \subset \mathbb{P}^n \text{ finite, reduced, and apolar to } F\bigr\}, \quad \text{(A.2)}$$

where X is apolar to F if and only if $I_X \subset F^{\perp}$, and $F^{\perp}$ denotes the apolar (Macaulay inverse) ideal of F. For $F = z^a$ this ideal is $F^{\perp} = (\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1})$, a complete intersection. The Hilbert function of the quotient $\mathbb{C}[\partial]/F^{\perp}$ then gives the lower bound matching (8); see [13, 14] for details.

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